高计 2 作业 2

截至日期：2026 年 4 月 3 日，中午 12 点

Total mark: 100

Part 1 (45 points): Critical Reading

In this exercise you will read the paper by **Kumar, and Liang (2024),** "Labor Market Effects of Credit Constraints: Evidence from a Natural Experiment". Answer the following questions.

Subpart 1: Background (15 points)

A. Define closed-end home equity loans (HELs) and home equity lines of credit (HELOCs) in Texas. What are the key similarities and differences between them? **(8 points)**

B. Figure 1 shows the decline in the Labor Force Participation Rate (LFPR) in Texas compared to the rest of the United States, with notable drops in 1997 and 2003. Based on Figure 1, can we infer the causal effect of the introduction of HELs and HELOCs in Texas? Explain why or why not. **(7 points)**

Subpart 2: Methodology (30 points)

C. Why do the authors use synthetic control method (SCM) in addition to the difference-in-differences (DID) method to estimate the causal effect? (Hint: Consider the advantages of SCM and the limitations of the DID method in this context.) **(12 points)**

D. When conducting SCM-ADH estimation, the authors observe that the quality of the match in pretreatment LFPR trends varies significantly across states, which could compromise the validity of the inference. How do the authors address this issue? **(12 points)**

E . What are the advantages of SCM based on machine learning (specifically SCM-Elastic Net and MC-NNM) compared to DID and SCM-ADH? Consider the restrictions each method imposes when the number of pre-treatment periods is small. **(6 points)**

Part 2 (55 points): Results Replication

A. **Replicate Figure 2**, which shows the pre-HEL (1992–1997) and post-HEL (1998–2007) LFPR path for Texas and synthetic Texas using the SCM-ADH specification with all pretreatment lags of LFPR to construct synthetic Texas. **(20 point)**

   a) The data required for replication is "state_year_panel_dataset.dta". The Stata

commands *synth* and *synth_runner* are helpful for replication.

b) You may run the *synth* command without rounding weights before conducting your formal synthetic control program. Please consult the "synth_no_rounding.ado" file provided for you.[1]

B. **Replicate Figure 3,** which plots the difference between LFPR paths of each placebo state and its synthetic control for the specification described in notes to Figure 2. **(15 point)**

C. **Replicate Figure 4,** which limited the donor pool to states that were similar to Texas in terms of major factors affecting the labor market in the posttreatment period: (i) energy states, (ii) states that did not change their minimum wage like Texas, (iii) states without state EITC, and (iv) states with similar welfare reform policies. The relevant variables are *energystate, noDminwage, DwelfareQ34, stateitc*. **(20 point)**

Please prepare a word document to briefly present your methods, and results required above. This word file will be the document for the marking process. In this file, include all the necessary tables and figures together with your explanations.

This word file **should not exceed 10 pages.**
If you use English, please use the **Times New Roman font, size 12, 1.5 line spacing**.
如果用中文答题和展示结果，请使用**楷体，12 号字体，1.5 倍行距。**

Name this word document as **yourname_homework2.doc/docx.** Submit the word file to course.pku.edu.cn before the deadline. **Any late submission will be subject to a 20% penalty on the final mark.**

[1] Note that the original synth program rounds weights after estimating those weights. To avoid rounding just use a version where rounding is commented out.